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# Load the dataset
from pandas import read_csv, read_excel
#url = 'https://raw.githubusercontent.com/jbrownlee/Datasets/master/
Longley.csv'
dataframe = read_excel('Simpleio.xlsx', header=None)
data = dataframe.values
# choose the input and output variables
xs, ys = data[:, 0], data[:, 1]
print('xs=',xs)
print('ys=',ys)

from statistics import mean
import numpy as np

#xs = np.array([1,2,3,4,5], dtype=np.float64)
#ys = np.array([5,4,6,5,6], dtype=np.float64)

def best_fit_slope(xs,ys):
    m = (((mean(xs)*mean(ys)) - mean(xs*ys)) /
          ((mean(xs)*mean(xs)) - mean(xs*xs)))
    return m

m = best_fit_slope(xs,ys)
print(m)

-0.6776137809227415

def best_fit_slope_and_intercept(xs,ys):
    m = (((mean(xs)*mean(ys)) - mean(xs*ys)) /
          ((mean(xs)*mean(xs)) - mean(xs*xs)))

    b = mean(ys) - m*mean(xs)

    return m, b
m, b = best_fit_slope_and_intercept(xs,ys)

print(m,b)

-0.6776137809227415 9.280489882016365

regression_line = [(m*x)+b for x in xs]
regression_line = []
for x in xs:
    regression_line.append((m*x)+b)

import matplotlib.pyplot as plt
from matplotlib import style
style.use('ggplot')
plt.scatter(xs,ys,color='#003F72')
plt.plot(xs, regression_line)
plt.show()

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predict_x = 7
predict_y = (m*predict_x)+b

plt.scatter(xs,ys,color='#003F72',label='data')
plt.plot(xs, regression_line, label='regression line')
plt.legend(loc=4)
plt.show()
```